Bias Bounty Mapping Equity Challenge

Discovery and methodology writeup

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Repository (code, reports, appendices): https://github.com/holimart/zindivias (CC-BY-SA 4.0)

Protocol A: V2_model_cgs — Zindi submission hvLtPVmr

Protocol B: B_from_A_cgs — Zindi submission eAaWXWxM

Form upload: PDF, maximum 10 MB (Zindi discussion #35334, 2026-10-14).

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Martin Holeček (Zindi: holi) — 2026-10-30

Executive summary

How to read this document: four findings follow, each opening with what was found and who it affects, then the evidence, then limits and a rebuild command. Throughout, *OSM share* means the share of a tract's Overture building footprints whose source list includes OpenStreetMap (tables also call it OSM-attributed or OSM-touched share); it records where footprints came from, not whether buildings are correct or complete. *pp* is percentage points; brackets hold 95% confidence intervals. Every number traces to a file in the public repository (Appendix G).

- **Rural source attribution:** rural tracts have 16.8–38.7 pp lower OSM-attributed building shares than urban tracts across all four regions (`reports/source-provenance.json`).
- **Rio Grande border metros:** mean OSM share is 0.092 vs 0.615 (border tracts vs the rest of South-Central Texas); the gap survives adjustment for Hispanic share and urbanity (β=–0.446), and 28.7% of border footprints carry a third `other` provider tag—the putative Google Open Buildings spill-over, a provider label, not a removed reference comparison (`reports/discovery-border-metros.json#/results`).
- **Paradise/Magalia:** in six burned tracts the Overture count is 0.620–0.747 of the provided WRC reference count, both within the Microsoft-derived WRC lineage; a fire-damage explanation was tested against CAL FIRE DINS inspections and not confirmed (0/6 focal passes; `reports/discovery-postfire.json#/focal_butte_2018_burn_geoids`; `reports/discovery-dins-mechanism.json#/decision`).
- **Tribal comparison:** the pooled result is region-confounded; 805 of 858 tribal tracts are in Eastern Oklahoma, so it largely compares regions, not tribal status (`reports/discovery-strata-confound.json#/stratum_composition`).
- **Out-of-region transfer:** the border pattern recurs in three unscored border counties; Google-tagged shares were 0.330 in El Paso, 0.292 in Imperial, and 0.108 in Doña Ana, versus 0.000 in both non-border controls (`reports/discovery-border-transfer.json#/results`).
- **Who is affected:** CDC PLACES 2025 estimates uninsured prevalence for adults aged 18–64 at 36.9% (about one adult in three) in the ten Texas border counties vs 18.1% elsewhere in the scored region (county-cluster 95% CI for the difference +16.6 to +22.1 pp); population context, not evidence about low-OSM tracts (`reports/who-is-affected-cdc-places.json`; `$COMPETITION_WORKSPACE/external-discovery-2026-10-29-fable/cdc-places-2025-tx/MANIFEST.json`).
- **Scored methods:** `V2_model_cgs` (a fixed-formula estimate from provided features) and its pre-registered `B_from_A_cgs` companion (the same estimate shrunk halfway toward its regional mean) use provided features only (`reports/final-two.json#/final_candidates`).

Submission context

- Protocol A: `V2_model_cgs`; leaderboard submission `hvLtPVmr`.
- Protocol B: `B_from_A_cgs`; leaderboard submission `eAaWXWxM`.
- Public RMSE (not private-split evidence): V2 0.1110; B 0.0975.
- Predictions use provided features only; no labels, hidden targets, placeholders, or removed references.
- All files cited by path (`reports/*.json`, `scripts/*`, `docs/writeup/appendices/*`) are in the submission repository: https://github.com/holimart/zindivias (CC-BY-SA 4.0); the organisers may also request it directly.

We request consideration for both Best Bias Discovery ($1,000) and Best Documentation ($500).

Best Bias Discovery entry

Finding 1 — Rural tracts have lower OSM-attributed shares of Overture buildings

Why it matters

What we found: in every scored region, rural buildings are far less often attributed to OpenStreetMap than urban buildings. Who it affects: rural residents in all four regions, including the high-vulnerability tracts named in Appendix I.

Across all four regions, rural tracts have lower mean OSM-attributed shares of Overture buildings than urban tracts. This is a provided-data source pattern, not an organizer-score comparison or evidence of missing or incorrect buildings.

Evidence

Read the “Rural − urban” column: negative in every region, with every interval excluding zero.

Region	Rural mean	Urban mean	Rural – urban (percentage points)	95% tract-bootstrap CI (pp)	n tracts (rural, urban)
Eastern Oklahoma	9.87%	26.70%	–16.83	[–19.24, –14.71]	1,191 (411, 780)

Region	Rural mean	Urban mean	Rural – urban (percentage points)	95% tract-bootstrap CI (pp)	n tracts (rural, urban)
Maricopa, Arizona	49.20%	87.38%	–38.17	[–43.15, –33.28]	1,582 (163, 1,419)
Northern California	24.70%	61.57%	–36.87	[–41.94, –32.01]	591 (147, 444)
South-Central Texas	25.15%	63.83%	–38.68	[–40.74, –36.56]	5,989 (991, 4,998)

Values are copied from `reports/source-provenance.json` (SHA-256 `e80171d07d555e8e77e533b35b8ae4ca71aee481e64f618d8a6dc63df0512565`). A tract is rural when the provided `pct_urban < 0.5` and urban when `>= 0.5`; tracts with missing `pct_urban` are excluded. Means are unweighted across tracts; intervals use 2,000 tract-bootstrap resamples. All CIs are below zero; no spatial adjustment was applied.

Named high-SVI examples

The report lists, per region, the ten tracts with the lowest non-null OSM-attributed share among the most vulnerable quarter of tracts (finite SVI, Q4); not all are rural. Appendix I gives their GEOIDs, county and place names, and shares; the names only annotate the preselected tracts. Appendix K shows OSM share across provided-SVI quartiles.

Rural, higher-vulnerability residents may rely on building context for heat, drought, wildfire, and evacuation planning; missing buildings, inaccurate maps, service harm, and dispatch/evacuation outcomes were not measured. Mappers should treat the named tracts as partner-led review priorities and verify proposed edits with local and tribal knowledge.

Limits and reproduction. OSM attribution is provenance: it does not measure accuracy, completeness, age, or local mapping activity, and one footprint may list several providers. Only provided Overture source fields and strata were used; no removed reference layers, scores, or candidate outputs (`reports/source-provenance.json`, cited SHA). Rebuild with `uv run --python 3.12 --no-project --isolated --with-requirements requirements.txt python scripts/discovery_source_provenance.py --workspace "$COMPETITION_WORKSPACE" --out reports/source-provenance.json --figure reports/source-provenance.png`.

Finding 2 — Rio Grande border metros: Overture's machine-only source mix

Impact and novelty

What we found: in the Rio Grande border metros, Overture's building footprints are overwhelmingly machine-sourced, and a third provider tag (read as Google Open Buildings) appears there but not in the non-border controls. Who it affects: the mostly urban residents of ten Texas border counties, where about one adult in three is uninsured; dispatch and evacuation outcomes were not measured.

Low OSM share alone is not the finding: the non-border control Ector has an even lower OSM share than El Paso (6.7% vs 9.3%) and 47.7% Esri, but zero share of the border provider tag. We measure source, not accuracy.

CDC PLACES 2025 estimates that 36.9% of adults aged 18–64 in these ten border counties are uninsured (about one adult in three), versus 18.1% elsewhere in the South-Central Texas scored region (difference +18.8 pp; county-cluster 95% CI +16.6 to +22.1 pp; 455 joined border tracts). This describes the border population where the place effect occurs; it is not evidence that low-OSM tracts are sicker. Health outcomes and dispatch/evacuation impacts were not measured (`reports/who-is-affected-cdc-places.json#/contrasts/ten border counties vs rest of South-Central Texas`).

Evidence and mechanism

Across the 460 border tracts the mean OSM share is 9.18%, against 61.48% elsewhere in the region (68.66% among urban tracts elsewhere); difference −52.30 pp (95% county-cluster CI −61.24 to −36.99). Adjusted for Hispanic share and urbanity, the border coefficient is β=−0.446 (SE 0.070; n=5,958, 185 counties); descriptive, not causal. County counts are in Appendix J.

`Other` is an Overture `sources` label, not removed Google Open Buildings or USGS reference data. By count matching, the border `other` count matches the first provider and the non-border Texas `other` count the second (both within ±526; Bexar cache share 0.336). This inference is supporting evidence, not proof (Appendix J). The Hispanic-share control comes from the CDC/ATSDR SVI layer in Appendix F.

Does it transfer?

We repeated the source-mix count in five counties outside the scored regions. “Google-only” denotes an Overture provider tag, not a removed reference layer: 10.8–33.0% of footprints in the three unscored border counties, zero in both non-border controls. El Paso's OSM and provider-tag shares (9.3%/33.0%) match the scored Rio Grande OSM range (4.2–12.0%); Doña Ana has 10.8% provider tag and 26.1% Esri. No interval is reported.

Footprint shares by source; the bold third-provider column is positive in the border counties and zero in both controls.

County	OSM	Third-provider tag	Esri
El Paso TX	0.093	**0.330**	0.000
Imperial CA	0.200	**0.292**	0.000

County	OSM	Third-provider tag	Esri
Doña Ana NM	0.179	**0.108**	0.261
Ector TX (control)	0.067	**0.000**	0.477
Bernalillo NM (control)	0.809	**0.000**	0.000

#35031/#35474 describe an Esri county data-sharing mechanism in Oklahoma; it coexists with ours: Esri is 47.7% in Ector and 26.1% in Doña Ana.

Full results and the rebuild command: Appendix J. Manifest: `$COMPETITION_WORKSPACE/external-discovery-2026-10-29-fable/MANIFEST.json`; rerun report `reports/discovery-border-transfer.json` (SHA-256 `64037e7360dab85eae3f748ec5d0ec14155999587302ae12e83bd3f9a893b795`). No scores or removed references.

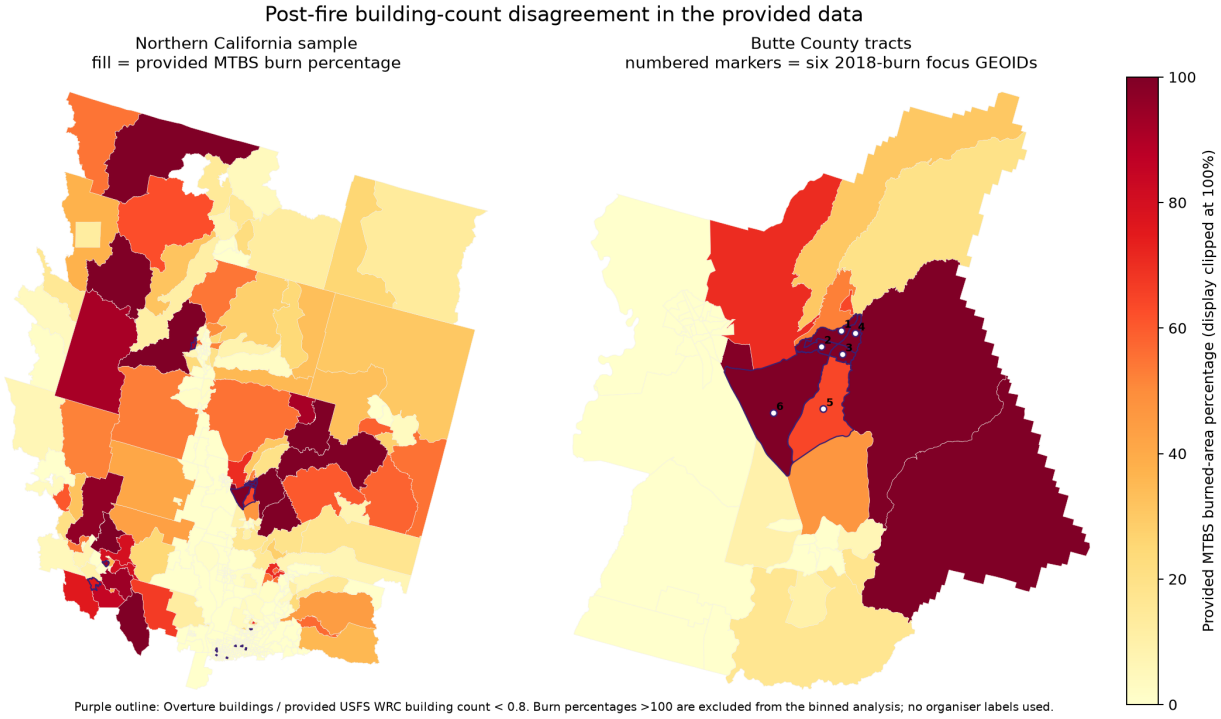
Finding 3 — Paradise/Magalia building-count discrepancy

Impact and evidence

What we found: in six burned Paradise/Magalia tracts the provided reference building count is well above Overture's, and both counts share a Microsoft-derived lineage, so the gap looks like a vintage difference within one provider, not missing OpenStreetMap mapping. Who it affects: planners who treat the larger count as current could misdirect field checks; we could not confirm Camp Fire damage as the cause.

In six Paradise/Magalia tracts (Butte County), the provided Wildfire Risk to Communities (WRC) building count exceeds the Overture count: Overture/WRC ratios are 0.620–0.747. This geographic and vintage discrepancy does not establish fire damage as its cause. The provided MTBS last-burn year is 2018. The public CAL FIRE Damage Inspection (DINS) dataset is used only as a discovery-stage mechanism check (retrieved 2026-10-29; catalogue CC BY, version unspecified).

Tract-level rows (focal GEOIDs, ratios and DINS counts) are in Appendix J (`reports/discovery-postfire.json#/focal_butte_2018_burn_geoids`; `reports/discovery-dins-mechanism.json#/focal_butte_tracts`). The tract-bootstrap difference is +9.55 pp (95% CI [+0.55, +20.89]), where the difference is the share of high-burn Northern California tracts (50–100% MTBS burned) whose Overture/WRC count ratio is below 0.8 minus the same share among unburned (0% burned) tracts; the county-cluster bootstrap 95% CI [−1.65, +16.67] includes zero, so the magnitude is exploratory (`reports/discovery-postfire-county-bootstrap.json#/county_cluster_bootstrap/cluster_95_percentile_interval`). The map marks these six tracts:



Map: fill is the provided MTBS burned percentage; the six focal tracts are marked.

Negative mechanism check. The DINS damage inspections support neither the fire-damage nor the rebuilding explanation: 0/6 focal tracts met the ≤0.30 relative-error rule (pass mark ≥4/6), and the Spearman correlation was 0.463 across 28 2018-burn tracts, below the 0.60 threshold. Among 591 NorCal tracts, 576 tracts with positive counts have no DINS record (1,139,778 buildings); a missing record does not establish absence or rebuilding. MTBS burned percentages are cumulative (64.4–153.8%); the reference count's vintage is unverified.

Why a vintage gap inside one provider: focal Microsoft-only shares are 0.430–0.990 versus OSM-only 0.001–0.089, and the Overture/WRC ratios of 0.620–0.747 are measured against a WRC count that is itself Microsoft-derived, so the gap is not OSM conflation (`reports/postfire-lineage-recheck.json#/focal_share_ranges`; `reports/discovery-postfire.json#/focal_butte_2018_burn_geoids`). Pooled OLS finds no OSM-only effect: β(OSM-only)=−0.016 (county-clustered SE 0.041; n=9,337; controls: urbanity, log density, region FE). Seven unburned NorCal tracts also have ratio <0.8, all urban (six Sacramento, one Placer; 0.711–0.775), so low ratios are not confined to burned tracts or to Sacramento (`reports/postfire-lineage-recheck.json#/pooled_ols`, `#/unburned_northern_ca_ratio_below_0_8`).

Entries #35520 and #35527 (23 Oct) provide stronger DINS-matched wildfire evidence; we acknowledge that work and limit Finding 3 here to the separate provider-lineage result and negative mechanism check (related-entry omissions ledger (related-entries.md)).

Reproduce with `scripts/discovery_postfire_stale_reference.py`, `scripts/audit_discovery_postfire_county_bootstrap.py`, and `scripts/discovery_dins_mechanism.py`; inputs, filters and hashes are recorded in reports/manifests.

Finding 4 — The tribal/non-tribal pooled comparison is region-confounded

Impact and novelty

What we found: the pooled tribal/non-tribal comparison mostly compares Eastern Oklahoma with Texas, because almost all tribal-flagged tracts are in Oklahoma. Who it affects: anyone reading the scorecard's tribal row as a general tribal disparity; compare within regions instead.

Regional composition can explain the pooled tribal/non-tribal contrast; this is not evidence of a general tribal disparity or causation.

Evidence

Of 858 `tribal_any` tracts, 805 (93.82%) are in Eastern Oklahoma; 6,002 of 8,521 non-tribal tracts (70.44%) are in South-Central Texas (`reports/discovery-strata-confound.json#/stratum_composition`).

What this does and does not show

These are composition counts, not equity or target-score evidence. `tribal_any` is an overlap indicator, not legal status; comparisons should be region-stratified (`reports/discovery-strata-confound.json#/regional_counts_and_prevalence`, `#/missing`).

Reproducibility

Rebuild: `python scripts/discovery_strata_confound.py --workspace "$COMPETITION_WORKSPACE" --out reports/discovery-strata-confound.json`.

Related entries on the challenge board

The author of #34744 (13 Sep) withdrew the colonia-gap claim on 7 Oct because it relied on removed reference layers; its replacement concerns “dispatchable roads” but gives no county, interval, or mechanism, while #35071 (28 Sep) uses a reconstructed target for its “colonias vacuum” and acknowledges tract labels do not verify colonia boundaries.

#34972 (23 Sep) and #35333 (14 Oct) describe rural road-vintage/phantom-road findings affecting roughly 33,000–400,000 people, not metropolitan building coverage, while #35011 (25 Sep) frames Hidalgo as a high-CVI tract set without a headline interval or Google-source analysis; our border-metro result instead covers 460 tracts and 1.77 million residents (89% urban) and reports an OSM-share association net of Hispanic share and urbanity (β=−0.446, SE=0.070, county-clustered).

#35031 (26 Sep) and #35474 (21 Oct) identify county data-sharing as a mechanism in Oklahoma but do not report Google Open Buildings or an out-of-region test; we add Google as a third provider in the border-building source mix and find Google-only footprints in all three out-of-region border counties (El Paso 33.0%, Imperial 29.2%, Doña Ana 10.8%) and in neither non-border control (Ector or Bernalillo, 0.0%).

#35264 (8 Oct) reports a pooled OSM-by-SVI gradient with leaderboard-calibrated gap magnitudes and regional-composition sensitivity, while #35030 (26 Sep) reports tribal-fire-station gaps in a sample where 805 of 858 tribal tracts are in Eastern Oklahoma; our San Antonio SVI contrast is within one city (Q4−Q1 −19.6 pp [−31.4, −7.9]) and our tribal comparison is explicitly region-confounded, not a general tribal effect.

#34995 (25 Sep) reuses the reconstructed-gap approach withdrawn in #34744 for its mobile-home result, #34962 (23 Sep) owns the address-layer finding (we make no address claim), and #35520/#35527 (23 Oct) provide stronger DINS-matched wildfire evidence; we cite that work and limit Finding 3 to the provider-lineage result (OSM-only 0.1–8.9%) and its negative mechanism check.

For who is affected, the public-domain CDC PLACES 2025 tract join (retrieved 29 Oct) describes the border population rather than low-OSM tracts: the ten border counties' uninsured prevalence for adults aged 18–64 is 36.9% versus 18.1% elsewhere in the region (county-clustered 95% CI for the difference +16.6 to +22.1 pp).

What ours adds
Border **metros + buildings**; Google as a third provider and cross-border spill-over; an adjusted place association net of Hispanic share; a San Antonio within-city gradient; an out-of-region transfer test; and a CDC PLACES health join about the border population.

Methodology (Best Documentation entry)

What the submitted score is

Two scored files were submitted, in this order. Protocol A (`V2_model_cgs`) is a fixed-formula estimate of each tract's coverage gap built only from the provided Overture and census features: it averages up to three component gaps (roads, buildings, points of interest), each clipped to the unit interval. Protocol B (`B_from_A_cgs`) takes A's score and pulls it halfway toward the mean of A over the tract's region. No labels, leaderboard scores, placeholders, or removed reference layers enter either file; the public scores are reported for transparency only.

Final-two: `V2_model_cgs` (`hvLtPVmr`; SHA-256 `5d0607d5531793008d8d5c86b1625f26e5bb8e9504a2b4a4a84f0224a0f5cef7`) and `B_from_A_cgs` (`eAaWXWxM`; SHA-256 `38e17cf1ec99f4f3fc5d358d13ccd0edfabe947d81fd6b6e5ef61e6cea006607`). Each is a 9,794-row `GEOID,coverage_gap_score` file copied byte-for-byte from its frozen source. 9,379 GEOIDs are official; the other 415 eastern-WA template rows use registered zero fill, not target estimates.

V2 components:

- **Transport:** `T_i=clip(1−named_road_km_5070_i/route_km_i,0,1)`, defined iff `route_km_i>0`; uses provided Overture named-road and eligible route lengths.
- **Buildings:** `G_i=0.01` iff the provided Overture building count is positive; otherwise undefined (placeholder zero excluded).
- **POI:** For fire/EMS/schools, `p_t=sigmoid(a_t+b_t·log1p((pop_total/max(ALAND_km²,.01))/100)+c_t·u)`, with fixed `(a,b,c)` respectively `(-2.8,.30,.70)`, `(-3.2,.25,.60)`, `(-1,.20,.35)`. `u=pct_urban` if finite, else `pop_urban/pop_total` if population positive, else `.5`. `n_t` is provided Overture category count plus its registered unclassified-place share. Type gap=`p_t×clip(1−n_t/(n_t+1),0,1)`, defined iff `p_t>0`; facility half averages defined types. Establishments half=`clip(1−all_overture_poi_count/(.024×pop_total),0,1)`, defined iff population or all-place count is positive (positive count at zero population gives zero). POI is the equal mean of defined halves, or undefined if neither is. Inputs: provided Overture category/all-place counts and `pop_total`, `pop_urban`, `pct_urban`, `ALAND`; priors and `.024` are fixed, not fitted to scores/labels.

V2 is the arithmetic mean of defined `T_i`, `G_i`, and `POI_i`; undefined terms are omitted. B changes only the composite: `B_i=.5×V_i+.5×mean(V_j)` over official tracts *j* in the same region. Its component columns are unchanged; its regional mean uses V2 predictions, not targets.

Fresh-clone rebuild: committed V2 source/components plus a workspace containing `samples/<region>-sample-submission.csv` (four official files) and `SampleSubmission.csv` (GEOID/region metadata) suffice; no private feature cache is needed. From the repository root:

```
uv run python scripts/build_candidate_b_from_a.py --workspace "$COMPETITION_WORKSPACE" && \
uv run python scripts/build_bias_column_order_cgs.py --workspace "$COMPETITION_WORKSPACE"
```

Verified in a scratch clone; both hashes matched.

Appendix A: Detailed methodology and estimator diagnostics

Detailed formulas, assumptions, and reproduction notes are in the linked methodology source (methodology.md). Alternative estimator-family and sensitivity diagnostics are in Appendix diagnostics (appendix-diagnostics.md).

Appendix B: Edge cases and documented limitations

The complete edge-case table and calculation policy are available in the linked edge-case appendix (edge-cases.md).

Appendix C: Screens that did not produce a finding

Supporting discovery screens and limitations are available in the linked screen appendix (appendices/screens-that-did-not-produce-a-finding.md). External-layer URLs, retrieval dates, and licence/terms status are summarized in Appendix F.

Appendix D: Leaderboard submission records and score eligibility

Historical five-column uploads placed `transport_gap` second, so their returned public values are preserved as records but void for A/B selection. The `_cgs` rows use `GEOID,coverage_gap_score`; the four cohort scores were used only by the frozen A rule, which selected the final A shown below; the B companion score was excluded from A selection. No public score is used to alter a candidate value. In the PDF each artifact SHA-256 is printed as four 16-character lines; the Markdown source of this appendix carries every hash unbroken.

Platform submission id	Candidate	Artifact	Artifact SHA-256	Public RMSE	Split	Score source	Selection eligibility	Uploaded (UTC)
uRRw9KpB	C0	`reports/candidates/C0_all_zero.csv`	`0edf8140123e2135489e5728fd71bce6f1e28a97f447ca411ba45fdbc02bc214`	0.107246299	public	zindi_api_status	void for selection: five-column file had transport_gap in column 2	2026-10-24T09:18:02+00:00
BvyKrR6v	C1	`reports/candidates/C1_region_constants.csv`	`dc2e368194b31408cb5c313fa8ce3f3fa59a1b5965bc2df8e4217bbc9e0c82e0`	0.146449112	public	zindi_api_status	void for selection: five-column file had transport_gap in column 2	2026-10-24T09:18:25+00:00
rTARuUaG	C2	`reports/candidates/C2_component_estimator_v0.csv`	`5728fbbaa2e062fbd14199190a260132c95b2b858da539d7a9580cb2abc62514`	0.235358712	public	zindi_api_status	void for selection: five-column file had transport_gap in column 2	2026-10-24T09:19:08+00:00
TM83xxU6	C3	`reports/candidates/C3_denominator_x125.csv`	`e5ee50f5c2cca149550f3408638d2b77431f6ed62cf104dfe8c07d8c8416f9e4`	0.349409867	public	zindi_api_status	void for selection: five-column file had transport_gap in column 2	2026-10-24T09:19:54+00:00
ySXi73rd	P4	`reports/candidates/P4_c1_shrink_alpha.csv`	`8b329802d882ea4bdff2bfd8c465c160d6ffd9269cfbe28dce8be9e702b07d37`	0.091440198	public	zindi_api_status	void for selection: five-column file had transport_gap in column 2	2026-10-24T09:38:39+00:00
iQw8C6Q9	A1_physical_equal	`reports/candidates/A1_physical_equal.csv`	`6af088b83ff3481d420e11b14d903fc510c2b4be2cea285c762ca12d500862de`	0.126543547	public	zindi_api_status	void for selection: five-column file had transport_gap in column 2	2026-10-26T14:23:12+00:00
W83LEghZ	A2_spectral	`reports/candidates/A2_spectral.csv`	`f88573996566a1975726845a49f42fb2cd3781d5f0e83d55944fc65647692997`	0.146089513	public	zindi_api_status	void for selection: five-column file had transport_gap in column 2	2026-10-26T14:23:12+00:00
tGB6sGRx	A3_dawid_skene	`reports/candidates/A3_dawid_skene.csv`	`6d588f2f7a45a58b128860a049ac19995b21902f7f98d3b2ab39d1d9fbb073b3`	0.146089513	public	zindi_api_status	void for selection: five-column file had transport_gap in column 2	2026-10-27T00:19:00+00:00
EbwLNkmt	V2_model	`reports/candidates/V2_model.csv`	`3193ee78a64004f007829f1f9c12d4aa8d4f7c15136771e8da9108fdcea06f12`	0.121644596	public	zindi_api_status	void for selection: five-column file had transport_gap in column 2	2026-10-27T00:19:00+00:00
hvLtPVmr	V2_model_cgs	`reports/candidates/V2_model_cgs.csv`	`5d0607d5531793008d8d5c86b1625f26e5bb8e9504a2b4a4a84f0224a0f5cef7`	0.11102814	public	zindi_api_status	selected A by frozen corrected-order _cgs rule	2026-10-28T07:15:46.098Z
4R4vkHSd	A1_physical_equal_cgs	`reports/candidates/A1_physical_equal_cgs.csv`	`2da0051af221ddf26a8a73740746811dbd956854a1fc12846c99864d3abae8a1`	0.309628459	public	zindi_api_status	corrected-order _cgs A-rule cohort; not selected as A	2026-10-28T07:15:51.544Z
s7mXS43p	A2_spectral_cgs	`reports/candidates/A2_spectral_cgs.csv`	`584e086bb5df950f6079317afcdc7c5a0182b5263d170be2a10987d1ac4c6213`	0.262578881	public	zindi_api_status	corrected-order _cgs A-rule cohort; not selected as A	2026-10-28T07:16:00.493Z
B8de77WD	A3_dawid_skene_cgs	`reports/candidates/A3_dawid_skene_cgs.csv`	`432ade8f1e46c6804cb34fb790612bb1befc9fe03b67a545e753be46b549237c`	0.15224784	public	zindi_api_status	corrected-order _cgs A-rule cohort; not selected as A	2026-10-29T00:10:29.255Z
eAaWXWxM	B_from_A_cgs	`reports/candidates/B_from_A_cgs.csv`	`38e17cf1ec99f4f3fc5d358d13ccd0edfabe947d81fd6b6e5ef61e6cea006607`	0.097535343	public	zindi_api_status	pre-registered B companion; score excluded from A selection	2026-10-29T00:12:18.199Z

The scored methodology candidate in this writeup is `V2_model`, built from provided features only (commit `b45733aa31903c71ec4f3fb432c2bc9d519f3ff9`). The original five-column V2 upload is void for selection because Zindi scored its transport_gap column. The frozen A rule selected V2_model_cgs from the four successful corrected-order _cgs scores (public RMSE 0.11102814; submission hvLtPVmr). B_from_A_cgs is its pre-registered 50% shrink companion (public RMSE 0.097535343; submission eAaWXWxM); B's score was not used to select A.

The earlier five-column uploads above are retained as historical calibration records for auditability. Their column-2 scores measured `transport_gap` and are void for selection; the correctly ordered `_cgs` evaluations are separately labeled, and no score value, fitted constant or probe outcome is used to change any candidate prediction or discovery claim.

Final-two protocol outcome: A=`V2_model_cgs` / `hvLtPVmr` (0.11102814); B=`B_from_A_cgs` / `eAaWXWxM` (0.097535343). A was selected from exactly four corrected-order `_cgs` scores; unique winner `V2_model_cgs` exceeded all pairwise noise guards. B is the pre-registered companion, not an A-rule competitor.

Appendix E: Prize consideration and official criteria

We request consideration for both Best Bias Discovery ($1,000) and Best Documentation ($500).

Best Bias Discovery entry — Best Bias Discovery ($1,000)

- Prize: Best Bias Discovery ($1,000).
- Official note: Entries for Best Bias Discovery are judged from the writeup by the organizing team against four criteria
- Judging criteria and weights (official text):
  - Impact: 30%.
  - Novelty: 25%.
  - Evidence: 25%.
  - Reproducibility: 20%.
- Tie-break: Ties are broken by earlier submission.

Best Documentation entry — Best Documentation ($500)

- Prize: Best Documentation ($500).
- Official note: No separate leaderboard submission is required for Best Bias Discovery or Best Documentation.

Appendix F: External layers and source register

Five public external layers were used only for the discovery screen described in the linked supporting analysis; none enters scored predictions. Retrieval dates and licence/terms verification status are recorded below. Where terms were not verified, no reuse or public-domain claim is made.

Layer	Source URL	Retrieved	Licence / terms status
CDC/ATSDR SVI 2022	https://svi.cdc.gov/Documents/Data/2022/csv/states/SVI_2022_US.csv	2026-10-27	Dataset-specific terms were not stated on the fetched ATSDR page. Treated as a U.S. federal government work under 17 U.S.C. Section 105, subject to third-party exceptions; this is a statutory assessment, not a quoted dataset-specific licence.
USDA ERS RUCA 2020	https://www.ers.usda.gov/media/5443/2020-rural-urban-commuting-area-codes-census-tracts.csv	2026-10-27	No terms/copyright statement found on fetched ERS page; unverified; ERS requests citation.
Census Planning Database 2026	https://www2.census.gov/adrm/PDB/2026/pdb2026tr.csv	2026-10-27	Not verified; Census terms/copyright pages were not read.
Census TIGER/Line 2020 AIANNH polygons	https://www2.census.gov/geo/tiger/TIGER2020/AIANNH/tl_2020_us_aiannh.zip	2026-10-27	Not verified; Census terms/copyright pages were not read.
CDC PLACES 2025, Texas census tracts	https://data.cdc.gov/500-Cities-Places/PLACES-Local-Data-for-Better-Health-Census-Tract-D/cwsq-ngmh	2026-10-29	Public Domain per Socrata metadata (license.name='Public Domain', licenseId='PUBLIC_DOMAIN'; U.S. federal work). Retrieved CSV SHA-256 and query/filter are recorded in $COMPETITION_WORKSPACE/external-discovery-2026-10-29-fable/cdc-places-2025-tx/MANIFEST.json; report SHA-256 and input hashes are in reports/who-is-affected-cdc-places.json.

Source-file hashes, download timestamps, and source-page licence checks are recorded in `$COMPETITION_WORKSPACE/external-discovery-2026-10-27/MANIFEST.json`; the data files are not redistributed in this packet.

Reproduce the CDC PLACES join and county-cluster intervals using `uv run --python 3.12 --no-project --isolated --with-requirements requirements.txt --with-requirements requirements-cdc-analysis.txt python scripts/audit_who_is_affected_cdc_places.py --workspace "$COMPETITION_WORKSPACE" --out reports/who-is-affected-cdc-places.json`. Input hashes, query/filter, outcomes, and limitations are recorded in the report and CDC workspace manifest.

The general register below lists prior-art sources, organiser-hosted challenge files, and public discovery endpoints. Scored predictions read only files in the organiser data package.

URL	Reference	Accessed	Role
https://api.zindi.world/v1/competitions/bias-bounty-mapping-equity-challenge/files/SampleSubmission.csv	organiser-hosted challenge file used by the workspace downloader	2026-10-23	official challenge data endpoint provided by the organisers; not an external dataset

URL	Reference	Accessed	Role
https://api.zindi.world/v1/discussions/35250	Organiser Challenge Update on supporting comparison evidence and target-reconstruction limits (discussion 35250)	2026-10-29	official rule clarification permitting discovery-only comparison evidence when not used to reconstruct, approximate, or calibrate targets
https://api.zindi.world/v1/discussions/35334	Zindi discussion 35334 (2026-10-14): methodology form requires PDF, maximum 10 MB	2026-10-29	organiser form upload-format instruction; this project exports a PDF under the stated size cap
https://zindi.africa/competitions/bias-bounty-mapping-equity-challenge	official challenge page (quoted rules and membership note)	2026-10-25	official challenge page; quoted text only, no data added to scored predictions
https://arxiv.org/abs/1502.04585	Citation: Avrim Blum and Moritz Hardt, “The Ladder: A Reliable Leaderboard for Machine Learning Competitions,” arXiv:150	2026-10-24	prior art and methodology know-how; not a dataset and never joined to scored predictions
https://arxiv.org/abs/1707.01825	Citation: Jacob Whitehill, “Climbing the Kaggle Leaderboard by Exploiting the Log-Loss Oracle,” arXiv:1707.01825. arXiv	2026-10-24	prior art and methodology know-how; not a dataset and never joined to scored predictions
https://data.ca.gov/api/3/action/package_show?id=cal-fire-damage-inspection-dins-data	California State Geoportal metadata record and CC BY catalogue terms	2026-10-29	public source metadata used to record DINS attribution licence and dataset limitations
https://data.ca.gov/dataset/cal-fire-damage-inspection-dins-data	CAL FIRE Damage Inspection (DINS) Data catalogue; public Best Bias Discovery source, CC BY version unspecified	2026-10-29	public external source used only for the Best Bias Discovery mechanism check; not joined to scored predictions
https://data.cdc.gov/500-Cities-Places/PLACES-Local-Data-for-Better-Health-Census-Tract-D/cwsq-ngmh	CDC PLACES 2025, Texas census tracts	2026-10-29	discovery-only population-health context for ten Texas border counties; not evidence about low-OSM tracts and never used in scored predictions
https://docs.overturemaps.org/schema/	Citation/documentation: Schema overview, Transportation Segment, and Places Place. URL check: all three official pages r	2026-10-24	prior art and methodology know-how; not a dataset and never joined to scored predictions
https://docs.overturemaps.org/schema/reference/places/place/	Citation/documentation: Schema overview, Transportation Segment, and Places Place. URL check: all three official pages r	2026-10-24	prior art and methodology know-how; not a dataset and never joined to scored predictions
https://docs.overturemaps.org/schema/reference/transportation/segment/	Citation/documentation: Schema overview, Transportation Segment, and Places Place. URL check: all three official pages r	2026-10-24	prior art and methodology know-how; not a dataset and never joined to scored predictions
https://doi.org/10.1016/j.buildenv.2023.110295	Citation: Biljecki et al. (2023), Building and Environment, article 110295, DOI 10.1016/j.buildenv.2023.110295; listed b	2026-10-24	prior art and methodology know-how; not a dataset and never joined to scored predictions
https://doi.org/10.1038/s41467-023-39698-6	Citation: Benjamin Herfort et al., “A spatio-temporal analysis investigating completeness and inequalities of global urb	2026-10-24	prior art and methodology know-how; not a dataset and never joined to scored predictions
https://doi.org/10.1080/13658816.2016.1189556	Citation: H. Senaratne et al. (2017), “A review of volunteered geographic information quality assessment methods,” Inter	2026-10-24	prior art and methodology know-how; not a dataset and never joined to scored predictions
https://doi.org/10.1080/13658816.2017.1395883	Citation: Zhou (2018), International Journal of Geographical Information Science, density–completeness study, DOI 10.108	2026-10-24	prior art and methodology know-how; not a dataset and never joined to scored predictions
https://doi.org/10.1111/tgis.12073	Citation: Christoph Barron, Pascal Neis, and Alexander Zipf (2014), “A comprehensive framework for intrinsic OpenStreetM	2026-10-24	prior art and methodology know-how; not a dataset and never joined to scored predictions
https://doi.org/10.1371/journal.pone.0180698	Citation: Christopher Barrington-Leigh and Adam Millard-Ball, “The world’s user-generated road map is more than 80% comp	2026-10-24	prior art and methodology know-how; not a dataset and never joined to scored predictions
https://doi.org/10.2737/RDS-2020-0016	Citation/documentation: Scott et al. (2020), Wildfire Risk to Communities: Spatial datasets of landscape-wide wildfire r	2026-10-24	prior art and methodology know-how; not a dataset and never joined to scored predictions
https://doi.org/10.3390/ijgi12040143	Citation: Ullah et al. (2023), ISPRS International Journal of Geo-Information 12(4), article 143, MapSwipe/completeness	2026-10-24	prior art and methodology know-how; not a dataset and never joined to scored predictions
https://doi.org/10.3390/ijgi2041066	Citation: Robert Hecht, Christian Kunze, and Stefan Hahmann, “Measuring Completeness of Building Footprints in OpenStree	2026-10-24	prior art and methodology know-how; not a dataset and never joined to scored predictions
https://doi.org/10.3390/ijgi7080289	Citation: Maria Antonia Brovelli and Giacomo Zamboni (2018), ISPRS International Journal of Geo-Information 7(8), articl	2026-10-24	prior art and methodology know-how; not a dataset and never joined to scored predictions
https://github.com/holimart/zindivias	public repository for the writeup and cited source files	2026-10-30	public submission repository; content shared under CC-BY-SA 4.0
https://github.com/microsoft/USBuildingFootprints	Citation/documentation: microsoft/USBuildingFootprints README. URL check: resolved; GitHub README fetched (repository is	2026-10-24	prior art and methodology know-how; not a dataset and never joined to scored predictions
https://overturemaps-us-west-2.s3.us-west-2.amazonaws.com/release/2026-09-23.1/theme=buildings/type=building/	This supporting transfer test uses Overture Maps Foundation buildings release 2026-09-23.1, the same release used by the	2026-10-24	prior art and methodology know-how; not a dataset and never joined to scored predictions

URL	Reference	Accessed	Role
https://services1.arcgis.com/iUJYIo9tSA7EHvfZ/arcgis/rest/services/POSTFIRE_MASTER_DATA_SHARE/FeatureServer/0	CAL FIRE DINS public point-feature service; filtered to Camp incident and non-address fields	2026-10-29	public external discovery data endpoint; not used in scored candidate construction
https://svi.cdc.gov/Documents/Data/2022/csv/states/SVI_2022_US.csv	CDC/ATSDR SVI 2022	2026-10-27	discovery-only external context; used for EP_HISP control in the border-metro descriptive regression; never used in scored predictions
https://uscode.house.gov/view.xhtml?req=granuleid:USC-prelim-title17-section105&num=0&edition=prelim	17 U.S.C. Section 105, statutory U.S. federal-work public-domain basis (subject to third-party exceptions)	2026-10-27	discovery-only external context; used for EP_HISP control in the border-metro descriptive regression; never used in scored predictions
https://www.census.gov/library/reference/code-lists/mt-feature-class-codes.html	Citation/documentation: U.S. Census Bureau, MAF/TIGER Feature Class Code Definitions. URL check: official Census page re	2026-10-24	prior art and methodology know-how; not a dataset and never joined to scored predictions
https://www.census.gov/programs-surveys/cbp/technical-documentation/methodology.html	Citation/documentation: U.S. Census Bureau, County Business Patterns Methodology. URL check: official Census page resolv	2026-10-24	prior art and methodology know-how; not a dataset and never joined to scored predictions
https://www.ers.usda.gov/media/5443/2020-rural-urban-commuting-area-codes-census-tracts.csv	USDA ERS RUCA 2020	2026-10-27	discovery-only external context; never used in scored predictions
https://www2.census.gov/adrm/PDB/2026/2026_PDB_Documentation.pdf	Census Planning Database 2026 documentation	2026-10-27	discovery-only external context; County_name used solely for naming preselected high-SVI tract examples; never used in scored predictions
https://www2.census.gov/adrm/PDB/2026/pdb2026tr.csv	Census Planning Database 2026	2026-10-27	discovery-only external context; County_name used solely for naming preselected high-SVI tract examples; never used in scored predictions
https://www2.census.gov/geo/tiger/TIGER2020/AIANNH/tl_2020_us_aiannh.zip	Census TIGER/Line 2020 AIANNH polygons	2026-10-27	discovery-only external context; never used in scored predictions
https://www2.census.gov/geo/tiger/TIGER2024/PLACE/tl_2024_04_place.zip	U.S. Census Bureau TIGER/Line 2024 place polygons, Arizona	2026-10-29	public domain Census geography used only to label preselected example tracts; no finding metric recomputed
https://www2.census.gov/geo/tiger/TIGER2024/PLACE/tl_2024_06_place.zip	U.S. Census Bureau TIGER/Line 2024 place polygons, California	2026-10-29	public domain Census geography used only to label preselected example tracts; no finding metric recomputed
https://www2.census.gov/geo/tiger/TIGER2024/PLACE/tl_2024_40_place.zip	U.S. Census Bureau TIGER/Line 2024 place polygons, Oklahoma	2026-10-29	public domain Census geography used only to label preselected example tracts; no finding metric recomputed
https://www2.census.gov/geo/tiger/TIGER2024/PLACE/tl_2024_48_place.zip	U.S. Census Bureau TIGER/Line 2024 place polygons, Texas	2026-10-29	public domain Census geography used only to label preselected example tracts; no finding metric recomputed
https://www2.census.gov/geo/tiger/TIGER2024/TRACT/tl_2024_04_tract.zip	U.S. Census Bureau TIGER/Line 2024 tract polygons, Arizona	2026-10-25	public domain Census geography used only to match preselected GEOIDs to Census places; no finding metric recomputed
https://www2.census.gov/geo/tiger/TIGER2024/TRACT/tl_2024_06_tract.zip	U.S. Census Bureau TIGER/Line 2024 tract polygons, California	2026-10-25	public-domain geography for outside-region footprint-to-tract assignment and matching preselected GEOIDs to Census places; not target reference data
https://www2.census.gov/geo/tiger/TIGER2024/TRACT/tl_2024_35_tract.zip	U.S. Census Bureau TIGER/Line 2024 tract polygons, New Mexico	2026-10-25	public-domain geography used only for outside-region footprint-to-tract assignment; not target reference data
https://www2.census.gov/geo/tiger/TIGER2024/TRACT/tl_2024_40_tract.zip	U.S. Census Bureau TIGER/Line 2024 tract polygons, Oklahoma	2026-10-25	public domain Census geography used only to match preselected GEOIDs to Census places; no finding metric recomputed
https://www2.census.gov/geo/tiger/TIGER2024/TRACT/tl_2024_48_tract.zip	U.S. Census Bureau TIGER/Line 2024 tract polygons, Texas	2026-10-25	public-domain geography for outside-region footprint-to-tract assignment and matching preselected GEOIDs to Census places; not target reference data

Provided-data-only check for the scored predictions (`reports/v2-input-provenance.json`): field-lineage fraction 1.0, 0 prohibited source dependencies, 0 unclassified inputs, external data added to scored predictions: False.

Appendix G: Reproducibility audit

A machine-checked claim-to-evidence audit backs every number in this writeup.

- `claims_total`: 83
- `claims_fully_traceable`: 83
- `fraction_quantitative_claims_traceable`: 1.0
- `numeric_tokens_total`: 971

- `numeric_tokens_covered`: 971
- `fraction_numeric_tokens_covered`: 1.0
- `distinct_verify_commands`: 83
- `distinct_reproduce_commands`: 21
- `derived_aggregates_verified`: 39
- `claims_asserting_accuracy`: 0

Current audit status: 83 of 83 quantitative claims are fully traceable (1.0); 971 of 971 numeric tokens are covered (1.0). There are 0 unresolved claim-evidence items. The full claim audit does not pass overall reproduction acceptance: 10 report-product differences remain outside accepted disclosure paths and 2 of 21 reproduction commands did not exit successfully (`reports/writeup-claim-evidence-audit.json#/commands`). This is a reproduction limitation, not an accuracy claim. The final-two result is separately traced to `reports/final-two.json` and machine-compared by the packet verifier.

The claim inventory, audit configuration, and execution details are available in `claim-evidence-audit.md` and the linked JSON reports: `reports/writeup-claim-evidence-audit.json`, `reports/writeup-claim-inventory.json`, and `reports/writeup-claim-evidence-execution.json`.

Deterministic audit artifacts: `reports/writeup-claim-evidence-audit.json`, `reports/writeup-claim-inventory.json`, `reports/writeup-claim-evidence-execution.json`.

Appendix H: Data provenance and prohibited-source statement

- Scored predictions are built only from the organiser-provided Overture roads, buildings and POI extracts, the provided census/ACS table, the provided strata table and the platform submission template, plus fixed documented priors.
- The organisers removed the reference layers (TIGER, MS Buildings, USGS, CBP). None of those layers, nor an equivalent copy, is used to build scored predictions or to reconstruct, derive, approximate, or calibrate target values.
- The organiser's 2026-10-08 Challenge Update (Zindi discussion #35250) permits removed reference layers as supporting comparison evidence in Best Bias Discovery, provided they are not used for target reconstruction or calibration. The Camp Fire comparison uses the provided USFS Wildfire Risk to Communities count and incident-filtered CAL FIRE DINS inspection points; the border finding uses a provided USFS building-count stratum. These are descriptive discovery comparisons only, not model inputs or target reconstruction. DINS provenance is recorded in Appendix F.
- Overture source tags named Google Open Buildings and USGS lidar are source labels in the provided source-provenance report; no external Google or USGS footprint/reference layer was accessed or used to build predictions or reconstruct the target.
- No sample score placeholder, no label, and no leaderboard-probe outcome is used as model evidence. The retired leaderboard-fitted constants and probe values of the retired probe set are listed in `reports/probe-retirement-2026-10-24.json` and appear nowhere in the model chain, in this writeup, or in any claim evidence, by name or by value.
- Equity and hazard strata columns are used only for the descriptive discovery screen, never as model inputs; the descriptive reports state that they are not accuracy measurements.
- All code and tooling used is open source, as the rules require.

Quoted official page text

Source: https://zindi.africa/competitions/bias-bounty-mapping-equity-challenge (documented 2026-10-25). official challenge page; quoted text only, no data added to scored predictions.

> *One membership note: the Maricopa package is 1,592 Arizona tracts plus one New Mexico tract, 35023970000 (Hidalgo County, Tract 9700), which the region boundary meets at the state line.*
>
> *Participants must not use reference layers, or equivalent copies obtained elsewhere, to recreate or approximate the organiser-computed targets.*

- Used for: cross-state scored tract disclosure in the edge-case section; independently verified present in the official Maricopa sample.
- Used for: prohibited-source statement.

Appendix I: Named high-SVI tracts with lowest OSM attribution

Selection and OSM-attributed shares below are copied from the 40 `named_lowest_osm_high_svi` report entries; no share or selection was recomputed. The list is the ten lowest non-null shares among finite-SVI Q4 tracts in each region, with Q4 defined within region. Source: `reports/source-provenance.json`, SHA-256 `e80171d07d555e8e77e533b35b8ae4ca71aee481e64f618d8a6dc63df0512565`.

County names are the Census PDB 2026 `County_name` field joined by `GEO_ID` (file SHA-256 `2faf8d075dfdf9841dc265c40b97f181a5716d604672eb130444138e363f65d5`). Place names are TIGER/Line 2024 `NAMELSAD` values from the Census place polygon with the greatest area overlap with each TIGER tract polygon, after projecting both to EPSG:5070; ties are alphabetical. An em dash means no place polygon overlaps. These geographic labels only annotate the preselected report records. The four state place-file URLs, retrieval dates, and manifest are listed in Appendix F and `$COMPETITION_WORKSPACE/external-discovery-2026-10-29/census-tiger-2024-places/MANIFEST.json`.

The table lists, per region, the ten most vulnerable (SVI Q4) tracts with the lowest OSM-attributed share; it is a review list for mappers, not a ranking of map errors.

Region	GEOID	County	Census place	OSM-attributed share
Eastern Oklahoma	40109108803	Oklahoma County	Oklahoma City city	0.063%
Eastern Oklahoma	40141070200	Tillman County	Tipton town	0.083%
Eastern Oklahoma	40109107103	Oklahoma County	Oklahoma City city	0.090%
Eastern Oklahoma	40107081000	Okfuskee County	Weleetka town	0.242%
Eastern Oklahoma	40101000300	Muskogee County	Muskogee city	0.281%
Eastern Oklahoma	40001376600	Adair County	Chewey CDP	0.316%
Eastern Oklahoma	40079040299	Le Flore County	Spiro town	0.411%
Eastern Oklahoma	40001376700	Adair County	Proctor CDP	0.434%
Eastern Oklahoma	40079040602	Le Flore County	Poteau city	0.460%
Eastern Oklahoma	40079040401	Le Flore County	Poteau city	0.498%
Maricopa, Arizona	04003000700	Cochise County	Douglas city	2.250%
Maricopa, Arizona	04021001305	Pinal County	Casa Grande city	2.803%
Maricopa, Arizona	04003000902	Cochise County	Douglas city	4.471%
Maricopa, Arizona	04021941401	Pinal County	Ak-Chin Village CDP	5.639%
Maricopa, Arizona	04007000900	Gila County	Claypool CDP	5.928%
Maricopa, Arizona	04013941200	Maricopa County	—	6.238%
Maricopa, Arizona	04021000904	Pinal County	Florence town	7.212%
Maricopa, Arizona	04003000600	Cochise County	Douglas city	7.212%
Maricopa, Arizona	04021001200	Pinal County	Coolidge city	7.820%
Maricopa, Arizona	04003000800	Cochise County	Douglas city	7.995%
Northern California	06007003200	Butte County	Palermo CDP	0.062%
Northern California	06115040301	Yuba County	Linda CDP	0.102%
Northern California	06007003300	Butte County	Palermo CDP	0.811%
Northern California	06033000501	Lake County	Lucerne CDP	1.339%
Northern California	06115040500	Yuba County	Olivehurst CDP	1.435%
Northern California	06007003100	Butte County	Oroville East CDP	2.065%
Northern California	06089012301	Shasta County	Redding city	2.122%
Northern California	06103000600	Tehama County	Red Bluff city	2.192%
Northern California	06115040901	Yuba County	Linda CDP	2.287%
Northern California	06103001000	Tehama County	Corning city	2.325%
South-Central Texas	48201433505	Harris County	Houston city	0.000%
South-Central Texas	48215020106	Hidalgo County	Mission city	0.000%
South-Central Texas	48215021315	Hidalgo County	Pharr city	0.000%
South-Central Texas	48215022011	Hidalgo County	South Alamo CDP	0.000%
South-Central Texas	48215022206	Hidalgo County	Midway North CDP	0.000%
South-Central Texas	48215022503	Hidalgo County	Olivarez CDP	0.000%
South-Central Texas	48215023520	Hidalgo County	San Carlos CDP	0.000%
South-Central Texas	48215023525	Hidalgo County	—	0.000%
South-Central Texas	48215024406	Hidalgo County	Olivarez CDP	0.000%
South-Central Texas	48479001822	Webb County	Laredo city	0.000%

Appendix J: Supporting discovery tables

Rio Grande border-county detail

Values are from `reports/discovery-border-metros.json`; Google attribution is regional, and provided-field Overture/USFS ratios are context, not Microsoft-target proxies. Look at the mean OSM share column: it stays low in the large counties (Hidalgo, Webb, Starr) while the other-source share is high.

Border county	Tracts	Population	Mean OSM share (95% tract CI)	Other-source share*	Median Overture / provided USFS count†
Hidalgo	212	870,781	6.10% (4.25–8.19%)	20.42%	1.04

Border county	Tracts	Population	Mean OSM share (95% tract CI)	Other-source share*	Median Overture / provided USFS count†
Cameron	121	421,017	13.27% (9.86–16.95%)	28.97%	1.39
Webb	69	267,114	6.16% (4.77–7.73%)	40.80%	1.56
Starr	17	65,920	1.91% (1.12–2.91%)	47.99%	1.75
Maverick	13	57,887	33.03% (21.53–44.33%)	61.25%	1.71
Val Verde	14	47,586	17.29% (6.78–33.24%)	13.85%	1.08
Zapata	6	13,889	27.63% (17.23–37.78%)	32.26%	1.42
Jim Hogg	2	4,838	3.35% (3.28–3.43%)	0.00%	0.98
Duval	3	9,831	3.71% (0.50–5.34%)	0.00%	1.02
Dimmit	3	8,615	3.51% (1.34–7.12%)	3.78%	0.63

*Weighted; Google attribution is regional. †Provided-field tract medians; small-county intervals are unstable.

Paradise/Magalia tract detail

The focal tract values are from `reports/discovery-postfire.json#/focal_butte_2018_burn_geoids`; DINS counts and relative errors are from `reports/discovery-dins-mechanism.json#/focal_butte_tracts`. Look at the last column: every relative error is far above the 0.30 pass threshold, which is why the fire-damage mechanism was not confirmed.

GEOID	Overture / reference	Reference – Overture	DINS destroyed	DINS major	Relative error
`06007001800`	0.620	666.4	3,314	3	0.799
`06007002100`	0.643	588.3	2,337	4	0.748
`06007002000`	0.662	579.5	2,870	4	0.798
`06007001900`	0.677	390.2	1,887	3	0.793
`06007002300`	0.720	668.9	2,386	2	0.720
`06007002200`	0.747	766.1	2,472	6	0.690

Out-of-region border transfer

This supporting transfer test uses Overture Maps Foundation buildings release 2026-09-23.1, the same release used by the lane cache. Dataset URL: `https://overturemaps-us-west-2.s3.us-west-2.amazonaws.com/release/2026-09-23.1/theme=buildings/type=building/`; retrieved 2026-10-29T20:36:09Z; Overture buildings ODbL 1.0 (Google Open Buildings CC BY 4.0/ODbL per Overture attribution). Downloaded by the STAC-pruned `overturemaps` CLI; the workspace manifest records exact bounding boxes, row counts and file hashes. The five counties are outside the four scored regions. Footprints are assigned by bbox centre to Census TIGER/Line 2024 county tracts solely for geography, not target reconstruction; these public-domain polygons are not the removed reference layer. Source classes use `scripts/estimate_gaps.py::classify_building_sources`. Shares are footprint-weighted; tract median OSM is unweighted; “tracts <5%” is a strict tract OSM-share threshold. Google-only means the single source name contains `google`.

County	Role	Footprints	OSM-touched	Microsoft-only	Google-only	Esri-touched	Tract-median OSM	Tracts <5% OSM / all
El Paso TX	Border, unscored	392,897	0.093	0.475	**0.330**	0.000	0.043	101 / 188
Imperial CA	Border, unscored	80,116	0.200	0.437	**0.292**	0.000	0.073	17 / 40
Doña Ana NM	Border, unscored	118,361	0.179	0.419	**0.108**	0.261	0.053	29 / 61
Ector TX (Odessa)	Non-border, Hispanic-majority control	78,050	0.067	0.456	**0.000**	0.477	0.036	18 / 33
Bernalillo NM (Albuquerque)	Non-border control	282,316	0.809	0.191	**0.000**	0.000	0.942	2 / 176

The lane `other` class is a mixed-provenance bin, not one provider: the border `other` count (254,346) is within 526 of Google Open Buildings memberships (254,872), while non-border South-Central Texas `other` (977,049) is within 526 of USGS lidar memberships (976,523). The Bexar provided-cache `other` share is 0.336. These count reconciliations support, but do not prove, the provider attribution; do not read `other` as Google statewide. Original source counts are in `reports/discovery-border-metros.json`; Bexar cache inputs and hash are recorded in `reports/discovery-border-transfer.json`.

Named impact context: Hidalgo tracts `48215021906`, `48215023525`, and `48215024404` have zero OSM-touched buildings and 1,727, 1,765, and 2,212 footprints; Cameron tract `48061012002` has 3,084 footprints and 0.03% OSM. Annual WBGT>30°C averages 53.5 days across 218 covered border tracts vs 7.3 across 3,654 covered non-border tracts; dispatch outcomes were not measured.

The source manifest at `$COMPETITION_WORKSPACE/external-discovery-2026-10-29-fable/MANIFEST.json` records the Overture public URL, CLI retrieval UTC, ODbL/attribution terms, bbox row counts and SHA-256 hashes. Reproduce the analysis with `uv run --python 3.12 --no-project --isolated --with-requirements requirements.txt python scripts/discovery/transfer_outside_regions.py --workspace "$COMPETITION_WORKSPACE" --out reports/discovery-border-transfer.json` (report SHA-256 `64037e7360dab85eae3f748ec5d0ec14155999587302ae12e83bd3f9a893b795`). External Census TIGER/Line 2024

tract files for CA, NM and TX were retrieved 2026-10-25 and are public-domain geography, not target reference data: https://www2.census.gov/geo/tiger/TIGER2024/TRACT/tl_2024_06_tract.zip, https://www2.census.gov/geo/tiger/TIGER2024/TRACT/tl_2024_35_tract.zip, https://www2.census.gov/geo/tiger/TIGER2024/TRACT/tl_2024_48_tract.zip. Archive SHA-256 values are in the report. The supporting test is `python -m unittest discover -s tests -p 'test_discovery_transfer_outside_regions.py' -v`. This discovery used public source provenance only; no candidate predictions or score values.

Appendix K: San Antonio-area SVI gradient

Evidence and method

This descriptive check uses only official sample GEOIDs, provided Overture building OSM-source share, and provided `svi_overall`, `svi_covered`, `pct_urban`, and population. Quartiles are computed within each region; the Bexar and urban-Texas subsets retain South-Central Texas cutpoints. Table values are equal-weight tract means. Q4−Q1 intervals are 5,000 independent tract-bootstrap resamples (seed 20261024). The measure is source attribution, not building accuracy or completeness.

Read the table left to right: within Texas, OSM share falls from the least vulnerable quartile (Q1) to the most vulnerable (Q4); the Q4−Q1 column gives the drop with its interval.

Scope	Q1	Q2	Q3	Q4	Q4–Q1, pp (95% CI)	n (Q1/Q4)
Eastern Oklahoma	26.8%	18.6%	18.2%	19.1%	–7.7 [–11.7, –3.8]	295/296
Arizona	84.8%	86.5%	80.8%	81.9%	–2.9 [–6.4, +0.6]	391/391
Northern California	55.6%	55.6%	50.6%	47.8%	–7.8 [–15.3, +0.1]	148/148
South-Central Texas	69.0%	57.0%	53.1%	50.3%	–18.8 [–21.4, –16.0]	1,490/1,490
Texas, urban (`pct_urban >= 0.5`)	73.5%	67.0%	62.1%	53.0%	–20.5 [–23.2, –17.8]	1,327/1,362
Bexar County, Texas	56.1%	65.9%	48.6%	36.5%	–19.6 [–31.4, –7.9]	55/139

Bexar County's 372 tracts show a higher-vulnerability tail: Q4 has 139 tracts and 653,292 residents; 33 Q4 tracts (142,547 residents) have <5% OSM-attributed share, versus 9 of 55 Q1 tracts. The 12 lowest-share Q4 GEOIDs, with shares from 0.30% to 1.70%, are `48029140600`, `48029160400`, `48029121506`, `48029160600`, `48029170700`, `48029140500`, `48029130300`, `48029151302`, `48029160502`, `48029171000`, `48029171301`, and `48029150700`. The Bexar SVI slope is −0.360 (HC1 SE 0.072); one county cannot support county-clustered uncertainty. The unadjusted Texas slope is −0.247 (county-clustered SE 0.103; 185 counties).

The Texas contrast is comparable in magnitude to the rural OSM-share contrast in Oklahoma, while Arizona's SVI interval crosses zero. This is a San Antonio-area pattern, not a cross-region rule. Source attribution does not establish map completeness, errors, causation, or effects on service delivery. No dispatch, evacuation, or relief outcomes were observed; use named tracts only as partner-led review priorities.

Reproduce: `uv run --python 3.12 --no-project --with-requirements requirements.txt python scripts/audit_san_antonio_svi_gradient.py --workspace "$COMPETITION_WORKSPACE" --out reports/san-antonio-svi-gradient.json`. The JSON report records exact estimates, bootstrap seeds, input SHA-256 values, scopes, and limitations.